

Campaign Analysis of company X

--total interaction per campaign (Which campaign generated the most engagement)

```
select c.campaign_name, count(ci.interaction_id) as interactions
from campaign_interaction ci
join campaigns c on
c.campaign_id = ci.campaign_id
group by c.campaign_name
order by interactions desc;
```

	campaign_name	interactions
1	Campaign_3	23
2	Campaign_5	23
3	Campaign_1	20
4	Campaign_4	18
5	Campaign_2	16

--total revenue by campaign_type (Which campaign made the most money)

```
select c.campaign_type, round(sum(cr.revenue_amount), 0) as total_revenue
from conversion cr
join campaigns c on c.campaign_id=cr.campaign_id
group by c.campaign_type
order by total_revenue desc;
```

	campaign_type	total_revenue
1	Discount	184351
2	Awareness	96893
3	Retargeting	70858

--conversion rate (Which campaign converts visitors into buyers most effectively)

```
select ci.campaign_id,  
count(distinct cr.conversion_id) * 100 / count(distinct ci.interaction_id) as conversion_rate  
from conversion cr  
left join campaign_interaction ci on ci.campaign_id=cr.campaign_id  
group by ci.campaign_id;
```

	campaign_id	conversion_rate
1	1	55
2	2	100
3	3	69
4	4	55
5	5	30

--clicks by campaign (campaigns generating most engagement)

```
select ci.campaign_id, c.campaign_type, count(interaction_id) as clicks  
from campaign_interaction ci  
left join campaigns c on c.campaign_id=ci.campaign_id  
where ci.interaction_type = 'Click'  
group by ci.campaign_id, c.campaign_type;
```

	campaign_id	campaign_type	clicks
1	2	Awareness	8
2	3	Discount	6
3	4	Discount	5
4	5	Discount	5
5	1	Retargeting	8

--calculate CTR

```
select ci.campaign_id, c.campaign_type,  
sum(case when ci.interaction_type = 'click' then 1 end) as clicks,  
sum(case when ci.interaction_type = 'view' then 1 end) as impressions,
```

```

sum(case when ci.interaction_type = 'click' then 1 end) * 100 /
sum(case when ci.interaction_type = 'view' then 1 end) as CTR
from campaign_interaction ci
left join campaigns c on c.campaign_id=ci.campaign_id
group by ci.campaign_id, c.campaign_type;

```

	campaign_id	campaign_type	clicks	impressions	CTR
1	2	Awareness	8	3	266
2	3	Discount	6	11	54
3	4	Discount	5	6	83
4	5	Discount	5	11	45
5	1	Retargeting	8	6	133

--channel generating highest number of conversions

```

select cu.acquisition_channel, count(distinct conversion_id) as conversions
from conversion cr
left join customers cu on cu.customer_id=cr.customer_id
group by cu.acquisition_channel
order by conversions desc;

```

	acquisition_channel	conversions
1	Organic	17
2	Google Ads	13
3	Facebook	12
4	Instagram	11
5	Referral	7

--customer segment analysis (segment of customers brings the most revenue)

```

select cu.customer_segment , round(sum(cr.revenue_amount), 0) as total_revenue
from conversion cr
left join customers cu on cu.customer_id=cr.customer_id
group by cu.customer_segment

```

order by total_revenue desc;

Results Messages		
	customer_segment	total_revenue
1	Returning	204952
2	VIP	103415
3	New	43735

--AOV by campaigns (which campaign attracts higher value customers)

```
select c.campaign_name, avg(cr.revenue_amount) as AOV
```

```
from conversion cr
```

```
join campaigns c on c.campaign_id=cr.campaign_id
```

```
group by c.campaign_name
```

```
order by AOV desc;
```

Results Messages		
	campaign_name	AOV
1	Campaign_1	6441.61636363636
2	Campaign_3	6388.62125
3	Campaign_2	6055.8275
4	Campaign_4	5155.865
5	Campaign_5	4367.77142857143

--ROI (which campaign deserves more budget)

```
select c.campaign_name, c.campaign_type, c.budget, sum(cr.revenue_amount) as revenue,
```

```
(sum(cr.revenue_amount)-c.budget)*100 / c.budget as ROI
```

```
from campaigns c
```

```
left join conversion cr on c.campaign_id=cr.campaign_id
```

```
group by c.campaign_name, c.campaign_type, c.budget;
```

Results Messages					
	campaign_name	campaign_type	budget	revenue	ROI
1	Campaign_1	Retargeting	154593.07	70857.78	-54.1649700080346
2	Campaign_2	Awareness	221020.45	96893.24	-56.1609615761799
3	Campaign_3	Discount	212436.53	102217.94	-51.88306832163
4	Campaign_4	Discount	67158.68	51558.65	-23.228613188943
5	Campaign_5	Discount	177801.63	30574.4	-82.8042071380336